

Macrosystems EDDIE: Getting Started + Troubleshooting Tips

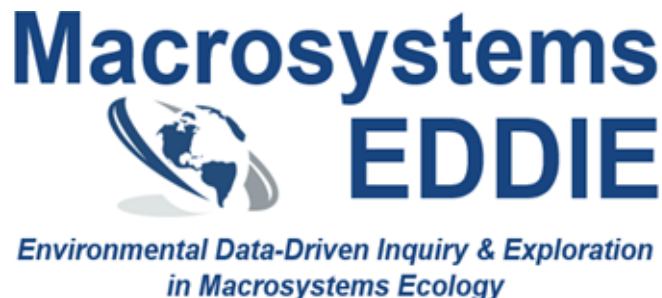
Lofton, M.E., Hipsey, M.H., Kurucz, K., and Carey, C.C. 08 August 2025.

Macrosystems EDDIE: Time Series Modeling and Prediction of Environmental Data.

Macrosystems EDDIE Module 11, Version 1.

<https://macrosystemseddie.shinyapps.io/module11/>

Module development supported by NSF EF 1702506, 1926050, 2318861, DEB 2213550, OISE 2330211



Swan-Canning Estuary Virtual Observatory



neon
Operated by Battelle

R Shiny Applications



- Statistical environment



- Interactive web app built using R.
 - Allows users to interact with data
 - Conduct their own analysis

Check-in:

- Can you access the Shiny app or this module?
 - Copy and paste this link into your browser:
<https://macrosystemseddie.shinyapps.io/module11/>
 - If this is not working contact us at MacrosystemsEDDIE@gmail.com and we will help you resolve this issue.

Landing Page of the Shiny App



Teaching materials associated with this module can be found at <http://module11.macrosystemseddie.org>.

Module 11: Time Series Modeling and Prediction of Environmental Data

🔖 Bookmark my Activity A progress

At any time, use this button to obtain a link that saves your progress in Activity A. Bookmarking for Activities B and C is not supported as these activities require users to upload their own data, which we do not store within the app. Please plan to allocate 45-60 minutes to complete Activities B and C.

Module Overview

Presentation

Introduction

Activity A

Activity B

Activity C

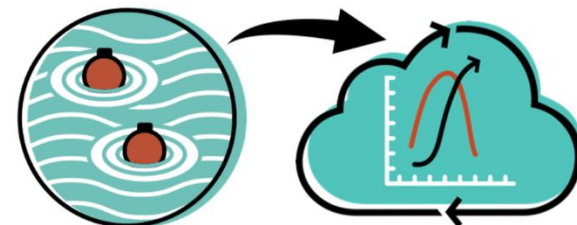
Time Series Modeling and Prediction of Environmental Data

Summary

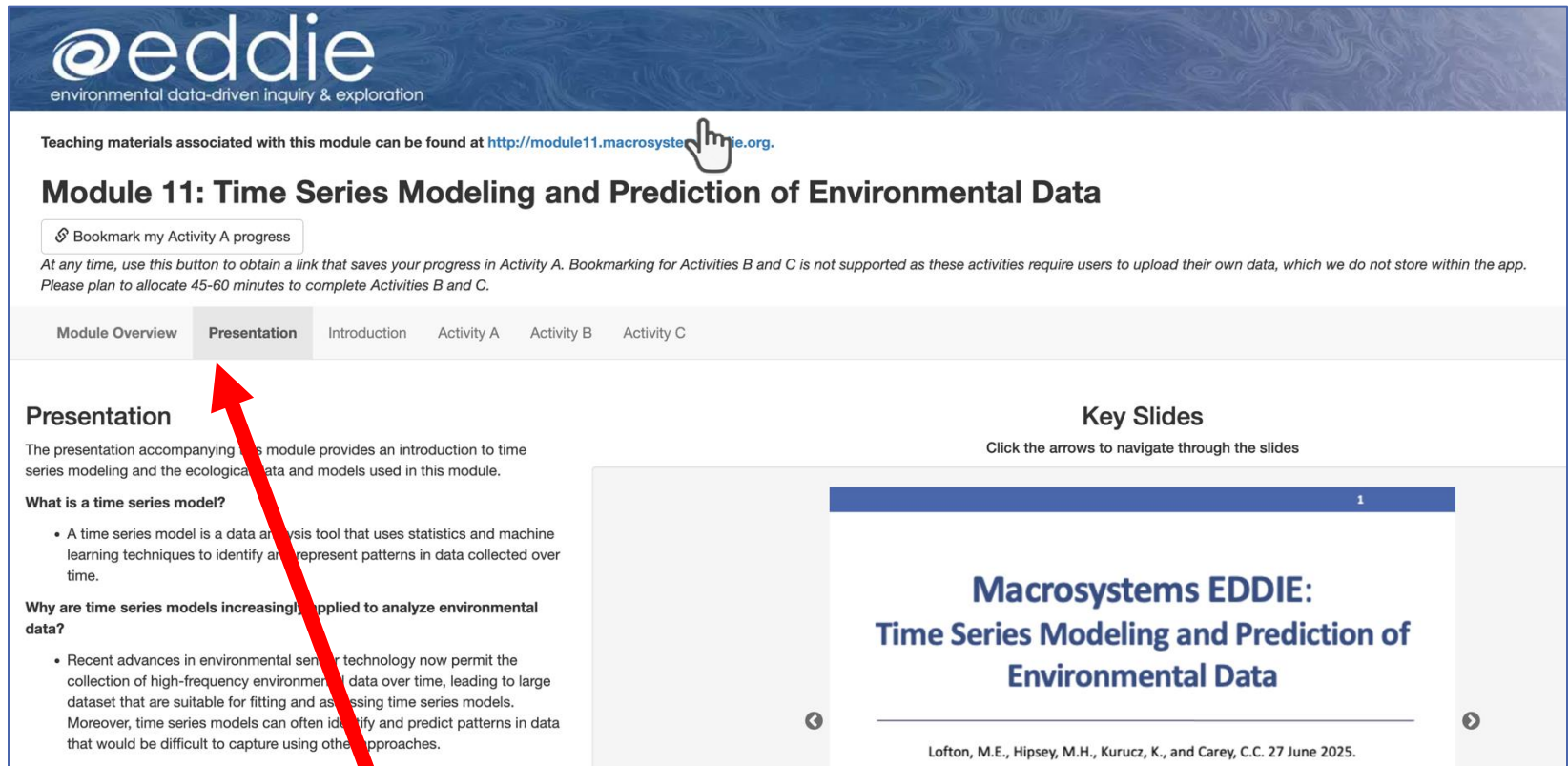
Advances in environmental sensor technology in recent decades permit collection of environmental data at high temporal frequencies (e.g., every 10 minutes) across many ecosystems. These time series of environmental data can be used to gain information about the previous and current state of ecosystems, as well as make predictions about ecosystem states in the future. To help researchers and managers understand the complex patterns that can occur in high-frequency time series data, time series models are commonly applied in ecology and environmental science. These models use statistical and machine learning methods to identify signals in high-frequency environmental data.

In this module, students will apply several different time series and machine learning models to environmental data. They will explore data from an environmental case study of their choice, choose which environmental variables to use to fit a time series model, assess the model, and use it to make predictions. Then, they will fit several other models to the same time series data and compare predictive performance across models. They will also evaluate the ecological implications that can be drawn from each model (e.g., which variables are important for explaining the dynamics of the target variable for prediction).

Time Series Modeling and Prediction of Environmental Data



Navigating the Shiny App



eddie
environmental data-driven inquiry & exploration

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Module 11: Time Series Modeling and Prediction of Environmental Data

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Module Overview **Presentation** Introduction Activity A Activity B Activity C

Presentation

The presentation accompanying this module provides an introduction to time series modeling and the ecological data and models used in this module.

What is a time series model?

- A time series model is a data analysis tool that uses statistics and machine learning techniques to identify and represent patterns in data collected over time.

Why are time series models increasingly applied to analyze environmental data?

- Recent advances in environmental sensor technology now permit the collection of high-frequency environmental data over time, leading to large datasets that are suitable for fitting and assessing time series models. Moreover, time series models can often identify and predict patterns in data that would be difficult to capture using other approaches.

Key Slides

Click the arrows to navigate through the slides

1

Macrosystems EDDIE: Time Series Modeling and Prediction of Environmental Data

Lofton, M.E., Hipsey, M.H., Kurucz, K., and Carey, C.C. 27 June 2025.

Select a tab by clicking on it

Navigate slides

Advance slides by clicking on the arrows

Presentation

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What is out-of-sample prediction?

- Out-of-sample prediction is using a model to make predictions for data that were not used to fit the model in the first place. So for example, if data from 2020-2024 are used to fit a model, using the model to make predictions for 2025 would be out-of-

Key Slides

Click the arrows to navigate through the slides



Interact with app

Objective 1 - Select case study

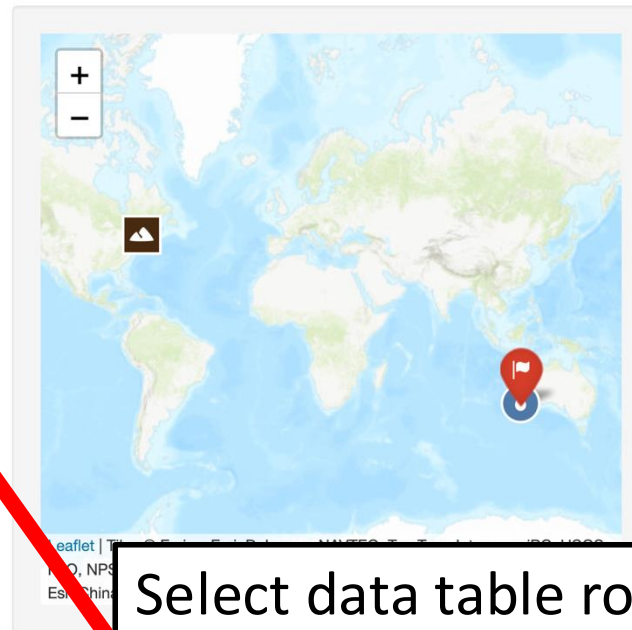
Select a case study from the table, then read the text in the 'About Case Study' section to find out more about the case study.

Site Names

Select a site in the table to highlight on the map

SiteID	SiteName	SiteLocation
1	cann	Canning River
2	bart	Bartlett Experimental Forest

Map of environmental case study sites



Site photo



credit: Mary Lofton

Select data table rows and click buttons

Interact with plots

Data Table

This is a Shiny data table. It is interactive and allows you to navigate through the data table by searching or clicking through the different pages.

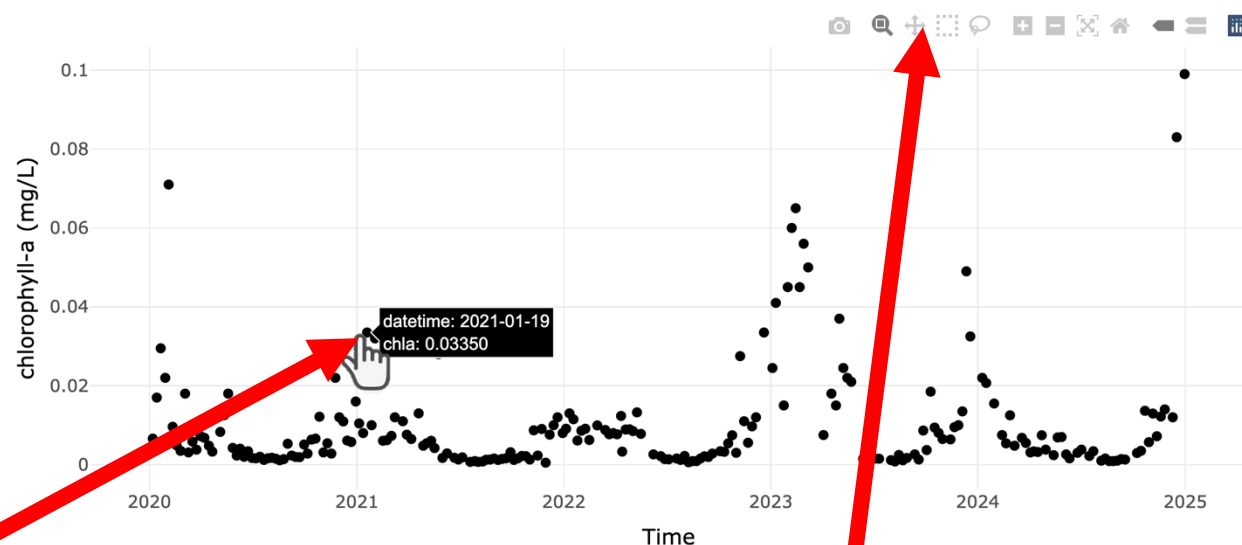
Show	10	entries	Search:	
	datetime	chl-a		
1	2020-01-07	0.0066		
2	2020-01-14	0.017		
3	2020-01-21	0.0295		
4	2020-01-29	0.022		
5	2020-02-04	0.071		
6	2020-02-11	0.0096		
7	2020-02-18	0.0049		
8	2020-02-25	0.0035		
9	2020-03-04	0.018		
10	2020-03-10	0.0031		

Data Plot

All plots in this Shiny app are generated using Plotly. This allows you to hover your mouse over the plot to get information from each of the plots. You can inspect the data closely by clicking and zooming into particular areas. There is also a tool box at the top of the plot.

Select variable

chlorophyll-a



Hover cursor over points or click and drag to zoom in

Hover cursor over plot to bring up options

Saving plots

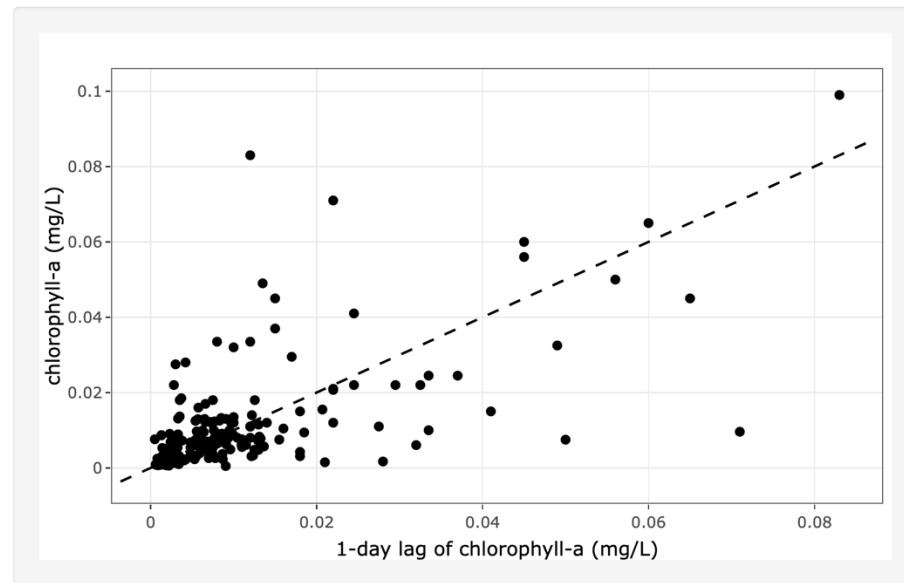
Plot 1-day lag of target variable

Let's explore lags and autocorrelation in the target data from your chosen environmental case study.

Plot lagged timeseries

Questions

Q8. Observe the plot on the right. Does the target variable exhibit autocorrelation at a 1-day lag? Explain how you know.



Download plot

Download plots to copy-paste into your final report

Downloading the Report

1. Navigate to the “Introduction” tab
2. Click on the “Download Final Report Template” button to download a Word document into which you can type your answers.

Student Handout

Within the Introduction and Activities A, B and C tabs there are questions for students to complete as part of this module. These can be completed by writing your answers into the final report template, which can be downloaded as a Word document (.docx) below.



Download Final Report Template



Answer questions

Type your answers into the final report template



Activity B: Upload your own data and fit a time series model

Objective 6: Upload standardized data

In this objective, you will learn about the importance of data standards for reproducible scientific analyses, as well as wrangle and upload a new environmental dataset of your choosing.

24. Describe one example of a benefit of using data standards in ecology and environmental science, as well as one example of a risk if data standards are not followed.

Answer:

25. Is the standardized data format for this module a long or a wide format? Explain how you know.

Answer:

26. Fill out the following information:

- a. Minimum number of timesteps required for dataset

Answer:

Saving & Resuming Progress

Saving Progress (Activity A only!)

1. Scroll to top of the page.
2. Click on the “Bookmark my progress” button. A pop-up window with a *very long link* will appear.
3. Copy-paste the link and store it at the top of your final report.



Teaching materials associated with this module can be found at <http://module11.macrosyste>

Module 11: Time Series Modeling and Predi

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Introduction

Activity A

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Activity C

Resuming progress

1. Open your browser.
2. Copy-paste the link into your browser.
3. As you navigate through the tabs in the module, your progress will reappear.

Bookmarked application link

```
https://macrosystemseddiedie.shinyapps.io/module6_dev/?
_inputs_&plot_airt_swt=0&plot_airt_swt2=0&plot_persist=0&plot_airt_swt3=0&plot_airt
_swt4=0&add_lm=0&add_lm1=0&plot_mlr=0&view_at_fc=0&fc1_Pers=0&fc1_Wtemp=
0&fc1_Atemp=0&fc1_Both=0&gen_proc_dist=0&fc2_Pers=0&fc2_Wtemp=0&fc2_Atem
p=0&fc2_Both=0&fit_model_year_1=0&fit_model_year_2=0&param_Pers=0&fc3_Pers=
0&param_Wtemp=0&fc3_Wtemp=0&param_Atemp=0&fc3_Atemp=0&param_Both=0&
fc3_Both=0&gen_ic=0&fc4_Pers=0&fc4_Wtemp=0&fc4_Atemp=0&fc4_Both=0&load_n
oaa_at=0&fc5_Pers=0&fc5_Wtemp=0&fc5_Atemp=0&fc5_Both=0&run_tot_fcA=0&qua
nt_ucA=0&run_tot_fcB=0&quant_ucB=0&dec_scen1=null&dec_scen2=null&values_&
sel_row=null&sel_mods=null
```

This link stores the current state of this application. Press ⌘-C to copy.

Dismiss

Progress-saving is not available for Activities B and C as the app does not store user data!

We recommend that you save your progress often!

- Because the Shiny app can time out after inactivity (15 minutes) or disconnect if an internet connection is interrupted, we don't want you to lose your work.
- Save your progress as you go, as well as every time you close your computer or close the Shiny app in your internet browser.
- After you save the link somewhere safe, you should be able to resume your progress where you left off!